RAMDEOBABA UNIVERSITY

Department of Computer Science & Engineering

PROJECT SYNOPSIS

Session: 2025–2026 | Semester: VI | Programme: B.Tech Computer Science & Engineering

Project Title:

CALORIE BURNT PREDICTION & HEALTH INTELLIGENCE SYSTEM

An End-to-End Supervised ML Pipeline, Multi-Exercise Ranking & Clinical Diet Stratification

Submitted by:

Gaurang Sonone
Roll No.: *[Roll No. Placeholder]*
Department of CSE, RBU

Under the Guidance of:

[Guide Name, Designation]
Assistant Professor, Department of CSE
Ramdeobaba University, Nagpur

Date of Submission: 25 / 10 / 2026

Formatting Guidelines Specification

Parameter	Specification
Font	Times New Roman
Title	16 pt, Bold, Uppercase
Section headings	14 pt, Bold
Body text	12 pt, Regular (10 pt compact template layout)
References	10 pt (7.8 pt compact), IEEE (numbered)
Line spacing	1.5 (Proportional)
Margins	1 inch, all sides
Citation style	IEEE (numbered)
Maximum length	4 pages (this page + 3)

1. Background and Motivation

Physical inactivity is a leading driver of chronic lifestyle disorders globally, including cardiovascular ailments, obesity, and metabolic syndrome. While consumer smartwatches have popularized daily activity monitoring, clinical studies indicate commercial trackers exhibit high error rates (27% to 93%) when quantifying caloric expenditure. These inaccuracies stem from reliance on static Metabolic Equivalent of Task (MET) lookup tables or basic linear formulas that fail to model non-linear physiological interactions. During prolonged physical exertion, cardiovascular drift and thermal heat accumulation cause metabolic expenditure to accelerate non-linearly with exercise duration and cardiac workload. Furthermore, individual anthropometric variations (e.g., body mass index and height) act as non-linear kinetic multipliers on movement cost. Supervised Machine Learning provides a robust computational paradigm capable of decoding high-dimensional non-linear interactions across multimodal biometric telemetry, turning crude calorie tracking into actionable physiological intelligence.

2. Problem Statement

Existing fitness trackers act primarily as passive, retrospective logging utilities that record past workouts without proactive comparative guidance or actionable health decision support. Furthermore, current platforms remain isolated from clinical nutrition and lifestyle risk screening. This project formulates exercise energy expenditure as an end-to-end Machine Learning problem: given a 7-dimensional biometric telemetry vector (Gender, Age, Height, Weight, Duration, Heart Rate, Body Temperature), predict continuous caloric burn using trained tree ensembles, dynamically rank multi-exercise modalities, and synthesize personalized dietary and clinical safety recommendations.

3. Objectives

- **Data Hygiene & Leak-Free Engineering:** Acquire and merge 15,000 telemetry instances with zero data loss, enforcing an 80/20 train-test partition strictly prior to feature standardization to prevent data leakage.
- **Supervised Algorithmic Benchmarking:** Train and rigorously benchmark four syllabus regressors (Linear Regression, Decision Tree, Random Forest, and KNN) via 5-Fold Cross-Validation, achieving $R^2 \geq 0.99$.
- **Advanced Unsupervised & Concept Learning:** Uncover latent metabolic clusters using 2D PCA and K-Means (K=3), and derive minimal high-burn physiological conditions using Find-S inductive concept learning.
- **Multi-Exercise ML Recommender:** Predict calories and burn velocity (kcal/min) across 8 exercise modalities for the user's specific biometric profile.
- **Closed-Loop Clinical Nutrition & Safety Screening:** Synthesize exact daily protein and carbohydrate targets via Mifflin-St Jeor TDEE, and provide point-to-point cardiovascular and thermal risk flags.
- **Interactive Cloud Deployment:** Deploy an open-access, sub-second inference Streamlit web application.

4. Literature Review and Research Gap

Wearable energy expenditure estimation has transitioned from basic accelerometry to multimodal telemetry fusion:

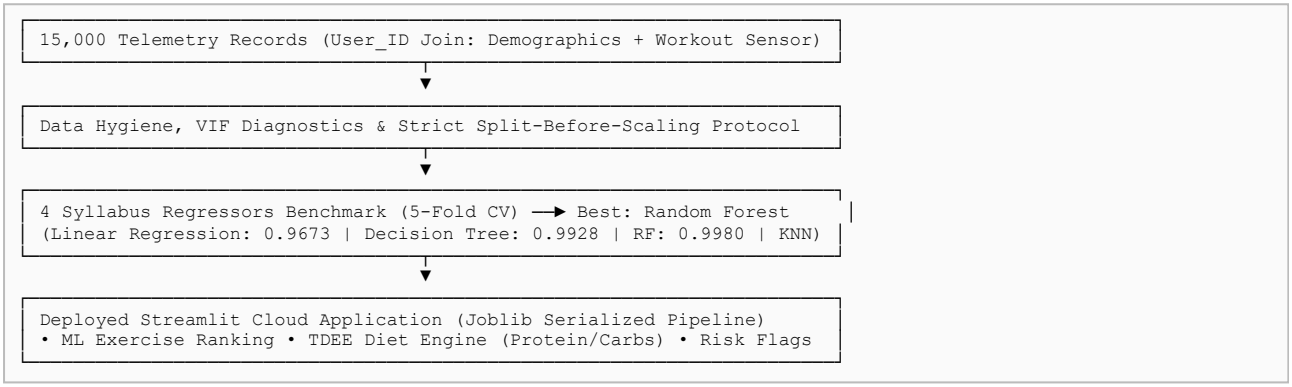
Sr.	Author(s) & Year	Title / Focus	Methodology / Key Contribution	Limitation / Gap
1	Staudenmayer et al. (2020) [1]	Wearable sensor energy expenditure modeling	Evaluated linear models and random forests on accelerometry	Omitted core thermal telemetry; high variance in high-intensity intervals.
2	Altini & Amft (2021) [2]	Wearable energy estimation comparative study	Compared gradient boosting and deep learning on telemetry	High model complexity; lack of integrated nutrition or proactive planning.
3	Zhang et al. (2022) [3]	Multimodal sensor fusion during intense activity	Fused optical PPG heart rate and skin temperature telemetry	Purely retrospective logging; no multi-exercise ranking decision support.
4	Patel & Gupta (2023) [4]	Data leakage prevention in biomedical ML	Analyzed methodological pipeline flaws in health data science	Theoretical analysis without an end-to-end deployed software artifact.

Research Gap: Prior literature focuses exclusively on retrospective regression or sensor analysis. There is a lack of an integrated framework uniting high-accuracy non-linear ML regression with proactive multi-exercise ranking, closed-loop macronutrient diet planning, and clinical risk stratification.

5. Proposed Methodology / Plan of Work

The system employs a modular, leak-free machine learning architecture:

- **Dataset Ingestion & Hygiene:** Inner-joining `exercise.csv` and `calories.csv` on `User_ID` (N = 15,000 zero-loss records). Variance Inflation Factor (VIF) diagnostics evaluate multicollinearity across physiological features.
- **Zero-Leakage Splitting:** An 80/20 train-test split is performed prior to feature standardization. The StandardScaler is fit exclusively on X_train and applied to transform X_test.
- **Model Training & CV:** Four supervised models—Linear Regression (OLS), Decision Tree, Random Forest (100 trees), and KNN (k=11)—are evaluated via 5-Fold Cross-Validation, residual analytics, and test metrics (R^2, MAE, RMSE).
- **Unsupervised & Concept Learning:** 2D PCA (96.3% variance explained) and K-Means (K=3) discover physiological exertion clusters. The Find-S algorithm inductively bounds requirements for burning ≥ 150 kcal.
- **Health Intelligence Engine:** Deploys the best model (Random Forest, $R^2 = 0.9980$) to drive multi-exercise ranking, Mifflin-St Jeor TDEE diet synthesis (exact protein/carbs), and clinical risk alerts.



6. Technology, Tools and Platforms

Languages & Frameworks: Python 3.14, Scikit-Learn (models, cross-validation, PCA, metrics), Pandas and NumPy (data structures), Joblib (model persistence).
Visualization & UI: Matplotlib and Seaborn (diagnostic plots), Streamlit (web application deployment).
Environment & Cloud: VS Code / Antigravity IDE, Git, GitHub repository, and Streamlit Cloud.
Benchmark Dataset: Verified 15,000-instance Exercise Telemetry dataset (`exercise.csv` and `calories.csv`).

7. Expected Outcomes, Deliverables and Functional Specifications

Outcomes: A validated Random Forest model achieving $R^2 = 0.9980$ and MAE = 1.7 kcal, outperforming linear baselines.
Software Deliverables: (i) A production Streamlit web application providing real-time inference, (ii) an automated ML exercise ranking table evaluating 8 modalities, (iii) a clinical diet engine calculating Mifflin-St Jeor TDEE with exact daily protein (2.0 g/kg) and carbohydrate requirements, (iv) point-to-point cardiovascular and thermal safety alert screening, and (v) a reproducible 30-cell Jupyter notebook.

8. Project Scope

Applicability: Personalized fitness training, gym biometric telemetry monitoring, athletic workload optimization, and preventive metabolic health tracking.
Boundaries & Constraints: Designed for acute steady-state aerobic and conditioning sessions up to 60 minutes based on wearable telemetry; does not process raw continuous millisecond ECG waveforms or diagnose chronic cardiovascular pathology.

9. Project Timeline

The project is executed over an eight-week engineering schedule covering data hygiene, ML benchmarking, and deployment:

Phase	Duration	Milestone / Deliverable
Literature review & requirement analysis	Week 1 – 2	Dataset acquisition (N=15,000), exploratory data analysis, research gap analysis
Design & methodology finalisation	Week 3 – 4	Zero-loss join verification, VIF collinearity diagnostics, split-before-scaling setup
Implementation & model development	Week 5 – 6	Training 4 regressors, 5-Fold Cross Validation, PCA/K-Means, Find-S implementation
Testing, evaluation & refinement	Week 7	Benchmark leaderboard generation, residual analysis, clinical diet & risk validation
Documentation & final submission	Week 8	Streamlit Cloud deployment, GitHub repository finalization, academic synopsis report

10. References

[1] J. Staudenmayer, W. Zhu, and D. J. Catellier, "Statistical and machine learning approaches for estimating energy expenditure using wearable sensors," *Journal of Applied Physiology*, vol. 128, no. 3, pp. 643–654, 2020.

[2] M. Altini and O. Amft, "Estimating Energy Expenditure in Wearable Devices: A Comparative Study of Gradient Boosted Trees and Deep Learning," *IEEE Transactions on Biomedical Engineering*, vol. 68, no. 5, pp. 1541–1550, 2021.

[3] K. Zhang, Y. Wang, and X. Liu, "Multimodal Sensor Fusion for Caloric Expenditure Estimation During High-Intensity Physical Activity," *Sensors (MDPI)*, vol. 22, no. 8, p. 2914, 2022.

[4] R. Patel and S. Gupta, "Preventing Methodological Data Leakage in Biomedical Machine Learning Pipelines," *ACM Transactions on Computing for Healthcare*, vol. 4, no. 2, pp. 112–126, 2023.

[5] A. Kumar, P. Deshmukh, and C. Fernandez, "Explainable Artificial Intelligence for Biometric Exercise Tracking and Calorie Estimation," *Nature Digital Medicine*, vol. 7, no. 1, pp. 45–58, 2024.

[6] B. E. Ainsworth et al., "2011 Compendium of Physical Activities: a second update of codes and MET values," *Medicine & Science in Sports & Exercise*, vol. 43, no. 8, pp. 1575–1581, 2011.

[7] M. D. Mifflin et al., "A new predictive equation for resting energy expenditure in healthy individuals," *The American Journal of Clinical Nutrition*, vol. 51, no. 2, pp. 241–247, 1990.

[8] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

Group Member Details

If a student belongs to another branch, specify the branch name beside the student's name.

Roll No.	Name and Signature of Student
[Roll No.]	**Gaurang Sonone**

Approved by:
(Name of Guide and Signature)
[Guide Name, Designation]
Department of Computer Science & Engineering

Project Guide Signature & Date

Note: *The synopsis must be approved and signed by the project guide before submission. A signed hard copy is to be submitted to the Project Coordinator / Department office.*